

Project 5 — Teaching a Webcam to Know Your Name

512 numbers and a name

A client wants a program that greets people by name when they sit down in front of a laptop webcam — and says “Unknown” for anyone it has never met. There is no giant database of faces to learn from and no time to train a network from scratch: the program has to learn a new person from a few seconds of webcam video, then recognize them live with a box, a name and a match score over every face in the frame. Adding a person has to happen inside the program itself — look at the camera, type a name, done. **Assume no cloud face-recognition service is allowed — every image and every stored face fingerprint stays on your own machine, only people who agree may be added, and the stored face data must be easy to delete.** Still recognizing yourself with glasses on, head turned or lights dimmed, while never mistaking the classmate next to you for yourself, is the harder half of the job — and it comes down to choosing one number and defending it. You are asked to vibe code this using InsightFace, and you have PyCharm and Python installed.

INSTRUCTIONS

Use the same structure for every project. Only the project details change.

PROJECT DEFINITION

PROJECT NAME

What are we building?

ONE-LINE GOAL

What should the project accomplish?

INPUTS

What information enters the system?

PROCESSING / “THE AI PART”

What should the program or AI do with the input?

OUTPUT / DEMO REQUIREMENT

What should happen when the project works?

IMPLEMENTATION GUIDANCE

HARDWARE IN THE LOOP

What hardware is involved, if any?

WHAT I WANT TO SEE WHILE IT RUNS

What should be visible on screen while the system operates?

TOOLS I HAVE / KNOW

What software, programming languages, AI tools or hardware do I have available?

CONSTRAINTS

What limitations or requirements must the project satisfy?

NOTES

- ◆ Ask clarifying questions first — especially hardware, privacy and consent, and tool choice.
- ◆ After agreement, phase the work: setup → input/data → build → visible → enroll/integrate → demo/test.
- ◆ Deliver PDF/Word/slides + an interactive checklist Artifact.

CODE

After we agree on the planning, generate one complete, runnable program whenever practical — covering enrollment, face matching and live webcam recognition in a single file. Keep it simple and beginner-friendly, with clear comments explaining important parts.